Chapter 10

Differentiation

In a bid to study the *rate of change* of a function, in this chapter we introduce the notion of the *derivative*. For instance, if $x : \mathbb{R} \to \mathbb{R}$ given by $x(t)$ describes the location of a car along a straight road at time t, then the rate of change of x gives the velocity of the car. The rate of change of the velocity is the acceleration.

10.1 The derivative

10.1.1 Definition and geometric meaning

To define the derivative, we start by defining a small increment in x and in y. For a function $f : \mathrm{dom}(f) \subseteq \mathbb{R} \to \mathbb{R}$, we fix $x_0 \in \mathrm{dom}(f)$ and assume that f is defined in a neighborhood $I_r(x_0)$ of x_0. Then for $x \in I_r(x_0)$ we denote

$$\Delta x = x - x_0$$

which is called the **increment in** x. Correspondingly we denote

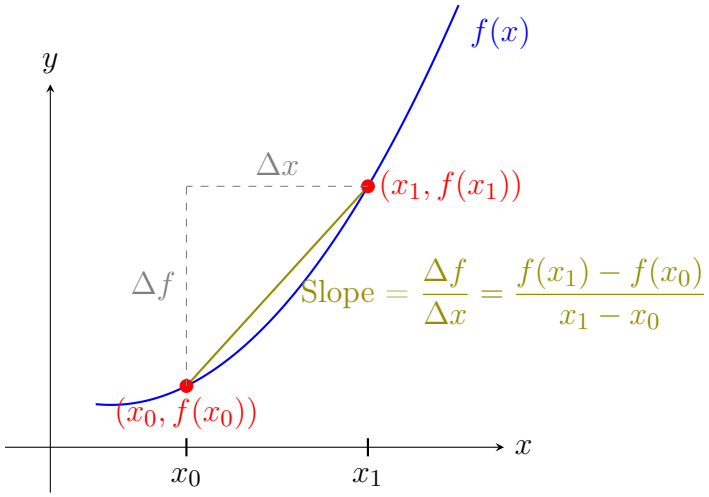
$$\Delta f = f(x) - f(x_0)$$

which is called the **increment in** f. We therefore have

$$x = x_0 + \Delta x \qquad \text{and} \qquad f(x) = f(x_0) + \Delta f.$$

The first step towards defining the derivative, is defining the **difference quotient of** f **between** x_0 **and** x:

$$\frac{\Delta f}{\Delta x} = \frac{f(x) - f(x_0)}{x - x_0} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$



The line connecting $(x_0, f(x_0))$ and another fixed point $(x_1, f(x_1))$ is called a **secant** and its equation is given by

$$s(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0)$$

By taking the limit of the secant as $x_1 \to x_0$ we obtain the *tangent line* to f at x_0, whose slope is the derivative of f at x_0:

Differentiability and the derivative

A function $f : \mathrm{dom}(f) \subseteq \mathbb{R} \to \mathbb{R}$ defined in a neighborhood of $x_0 \in \mathrm{dom}(f)$ is called **differentiable at** x_0 if $\lim_{x \to x_0} \frac{\Delta f}{\Delta x}$ exists and is finite. This number is called the **derivative of** f **at** x_0 and is denoted by $f'(x_0)$:

$$f'(x_0) = \lim_{x \to x_0} \frac{\Delta f}{\Delta x} = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

Other symbols used to denote the derivative at x_0 include

$$\frac{\mathrm{d}f}{\mathrm{d}x}(x_0) \qquad \dot{f}(x_0) \qquad \mathrm{D}f(x_0)$$

Given $I \subseteq \mathrm{dom}(f)$, we say that f **is differentiable on** I if f is differentiable at each $x_0 \in I$.

Newton, Leibniz, and who invented calculus

The many notations above reflect a turbulent history. Isaac Newton (1643–1727) developed his "method of fluxions" in 1665–66, but did not publish it for decades. Gottfried Wilhelm Leibniz (1646–1716), then living in Paris, independently developed his own version of calculus in 1675, and published it first, in 1684. What followed was one of the most bitter priority disputes in the history of science. In 1711 John Keill accused Leibniz, in the journal of the Royal Society, of plagiarising Newton. When Leibniz appealed to the Society, it set up a committee that never asked for his side of the story, and whose report (1713) was in fact written by Newton himself, the Society's president. Leibniz died in 1716 with the matter unresolved; according to an eyewitness, his only mourner was his secretary. Today historians agree that the two discovered calculus independently. Their notations fared differently: Newton's dots ($\dot{y}$) survive mainly in physics, for derivatives with respect to time, while Leibniz's $\frac{\mathrm{d}y}{\mathrm{d}x}$ – together with his integral sign $\int$, an elongated S for summa *– is used throughout mathematics. (The prime notation f', usually named after Lagrange, was in fact introduced by Euler.)*

We observe that this defines a new function:

The derivative function

The definition of the derivative at a point induces the definition of the **derivative function** f' whose value at x_0 is $f'(x_0)$. Its domain is

$$\mathrm{dom}(f') = \{x \in \mathrm{dom}(f) \quad | \quad f \text{ is differentiable at } x\}.$$

10.1.2 Differentiability and continuity

Proposition 10.1: If f is differentiable at x_0 it is also continuous at x_0.

Proof. We have

$$\begin{aligned}\lim_{x\to x_0}(f(x)-f(x_0)) &= \lim_{x\to x_0}\frac{f(x)-f(x_0)}{x-x_0}(x-x_0)\\ &= \lim_{x\to x_0}\frac{f(x)-f(x_0)}{x-x_0}\lim_{x\to x_0}(x-x_0)\\ &= f'(x_0)\cdot 0 = 0.\end{aligned}$$

□

Remark: The other implication is not true. For instance, the function $f(x) = |x|$ is continuous on $\mathbb{R}$, and in particular at $x_0 = 0$, however it is *not differentiable* at $x_0 = 0$. This can easily be seen by taking limits from the left and from the right and seeing that

they are different:

$$\lim_{x\to x_0^+}\frac{f(x)-f(x_0)}{x-x_0}=\lim_{x\to 0^+}\frac{x}{x}=1,$$
$$\lim_{x\to x_0^-}\frac{f(x)-f(x_0)}{x-x_0}=\lim_{x\to 0^-}\frac{-x}{x}=-1.$$

Linear approximation

Let f be differentiable at x_0. Then

$$f(x)=f(x_0)+f'(x_0)(x-x_0)+o(x-x_0),\qquad x\to x_0,$$

or, writing $\Delta x=x-x_0$ and $\Delta f=f(x_0+\Delta x)-f(x_0)$,

$$\Delta f=f'(x_0)\Delta x+o(\Delta x),\qquad \Delta x\to 0.$$

Conversely, if $f(x)=f(x_0)+m(x-x_0)+o(x-x_0)$ as $x\to x_0$ for some $m\in\mathbb{R}$, then f is differentiable at x_0 and $f'(x_0)=m$. (In some textbooks this is called the *first increment formula*.)

Indeed, by the definition of the derivative,

$$\lim_{x\to x_0}\frac{f(x)-f(x_0)-f'(x_0)(x-x_0)}{x-x_0}=\lim_{x\to x_0}\frac{f(x)-f(x_0)}{x-x_0}-f'(x_0)=0,$$

which says exactly that the numerator is $o(x-x_0)$; the converse is the same computation read backwards. So near x_0 the graph of f is approximated by its tangent line $y=f(x_0)+f'(x_0)(x-x_0)$, with an error that is small compared with $x-x_0$ (see Figure 10.1).

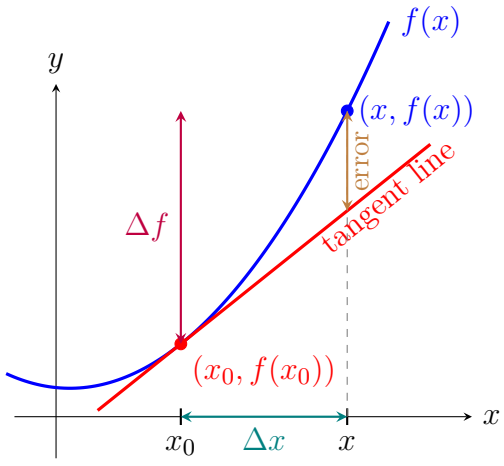


Figure 10.1: The 'error' is of order $o(\Delta x)$ as $\Delta x\to 0$

10.1.3 Derivatives of elementary functions

Example 10.1: Consider the affine function $f(x) = ax + b$. Then:

$$f'(x_0) = \lim_{x\to x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x\to x_0} \frac{ax + b - ax_0 - b}{x - x_0} = a \lim_{x\to x_0} \frac{x - x_0}{x - x_0} = a.$$

Since this is true for all $x_0 \in \mathbb{R}$, we find that the derivative function of $f(x)$ is $f'(x) = a$.

Example 10.2: Consider the quadratic function $f(x) = ax^2$:

$$f'(x_0) = \lim_{x\to x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x\to x_0} \frac{ax^2 - ax_0^2}{x - x_0} = a \lim_{x\to x_0} \frac{(x - x_0)(x + x_0)}{x - x_0} = 2ax_0.$$

Since this is true for all $x_0 \in \mathbb{R}$, we find that the derivative function of $f(x)$ is $f'(x) = 2ax$.

Example 10.3: Consider a monomial $f(x) = ax^n$, $n \in \mathbb{N}_+$. Using the formula

$$u^n - v^n = (u-v)\left(u^{n-1} + u^{n-2}v + u^{n-3}v^2 + \cdots + uv^{n-2} + v^{n-1}\right) = (u-v)\sum_{k=0}^{n-1} u^k v^{n-1-k}$$

we have

$$\begin{aligned} f'(x_0) &= \lim_{x\to x_0} \frac{f(x) - f(x_0)}{x - x_0} \\ &= \lim_{x\to x_0} \frac{ax^n - ax_0^n}{x - x_0} \\ &= a \lim_{x\to x_0} \frac{(x - x_0)\sum_{k=0}^{n-1} x^k x_0^{n-1-k}}{x - x_0} \\ &= anx_0^{n-1}. \end{aligned}$$

Since this is true for all $x_0 \in \mathbb{R}$, we find that the derivative function of $f(x)$ is $f'(x) = anx^{n-1}$.

Example 10.4: We can also consider the more general power function $f(x) = x^\alpha$, $\alpha \in \mathbb{R}$. Here we will use the identity (proved in Section 6.7)

$$\lim_{y\to 0} \frac{(1+y)^\alpha - 1}{y} = \alpha, \qquad (\alpha \in \mathbb{R}).$$

Let $x_0 > 0$. Then

$$\begin{aligned} f'(x_0) &= \lim_{x\to x_0} \frac{f(x) - f(x_0)}{x - x_0} \\ &= \lim_{\Delta x\to 0} \frac{(x_0 + \Delta x)^\alpha - x_0^\alpha}{\Delta x} \\ &= x_0^{\alpha-1} \lim_{\Delta x\to 0} \frac{(1 + \frac{\Delta x}{x_0})^\alpha - 1}{\frac{\Delta x}{x_0}} \\ \left(\textit{substitute } y = \frac{\Delta x}{x_0}\right) \quad &= x_0^{\alpha-1} \lim_{y\to 0} \frac{(1+y)^\alpha - 1}{y} \\ &= \alpha x_0^{\alpha-1}. \end{aligned}$$

This shows that $f'(x) = \alpha x^{\alpha-1}$ for all $x > 0$. (We divided by x_0 and wrote $(x_0 + \Delta x)^\alpha = x_0^\alpha \left(1 + \frac{\Delta x}{x_0}\right)^\alpha$, which is why we need $x_0 > 0$.) For $\alpha = n \in \mathbb{N}$, the previous example gives the same formula for all $x \in \mathbb{R}$.

Example 10.5: The function

$$f(x) = \sqrt{x} = x^{\frac{1}{2}} \quad \text{has domain } \mathrm{dom}(f) = [0, +\infty)$$

while its derivative

$$f'(x) = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}} \quad \text{has domain } \mathrm{dom}(f') = (0, +\infty).$$

Example 10.6: For the function $f(x) = \sin x$ we use the formula

$$\sin(\alpha) - \sin(\beta) = 2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

as follows:

$$\begin{aligned}
f'(x_0) &= \lim_{x\to x_0} \frac{f(x) - f(x_0)}{x - x_0} \\
&= \lim_{x\to x_0} \frac{\sin x - \sin x_0}{x - x_0} \\
&= \lim_{x\to x_0} \frac{2\cos\left(\frac{x+x_0}{2}\right)\sin\left(\frac{x-x_0}{2}\right)}{x - x_0} \\
&= \underbrace{\lim_{x\to x_0} \cos\left(\frac{x+x_0}{2}\right)}_{=\cos x_0} \underbrace{\lim_{x\to x_0} \frac{\sin\left(\frac{x-x_0}{2}\right)}{\frac{x-x_0}{2}}}_{=1} \\
&= \cos x_0.
\end{aligned}$$

Since this is true for all $x_0 \in \mathbb{R}$, we find that the derivative function of $f(x)$ is $f'(x) = \cos x$.

Example 10.7: Similarly it can be shown that for $f(x) = \cos x$, $f'(x) = -\sin x$.

Example 10.8: Let $f(x) = a^x$. Then

$$\begin{aligned}
f'(x_0) &= \lim_{\Delta x\to 0} \frac{a^{x_0+\Delta x} - a^{x_0}}{\Delta x} \\
&= a^{x_0} \underbrace{\lim_{\Delta x\to 0} \frac{a^{\Delta x} - 1}{\Delta x}}_{=\ln a} \\
&= a^{x_0} \ln a
\end{aligned}$$

Since this is true for all $x_0 \in \mathbb{R}$, we find that the derivative function of $f(x)$ is $f'(x) = a^x \ln a$. In particular,

$$(e^x)' = e^x.$$

The functions ce^x ($c \in \mathbb{R}$) are all equal to their own derivative, and we will see later that they are the only ones. This is one reason why the number e is so special.

Example 10.9: The area A of a disc of radius r is given by πr^2. Thinking of A as being a function of r, we can write

$$A(r) = \pi r^2.$$